

BSc (Hons) in Information Technology – Year 3

IT3060 – Human Computer Interaction

Semester 2, 2026

Lab Exercise 04

Usability Testing for the FitFlow App Redesign

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Date of Submission: 2026.08.30

Activity 1: Define Usability Goals

Usability goals were set to be specific and measurable, and were explicitly tied to the business problems that motivated the redesign in the first place: declining retention, falling app-store ratings, and the pain points already validated in Labs 1–2 (limited personalization, cumbersome logging, poor progress visualization).

1.1 Goals Linked to Business Problems

Business problem	Usability goal	Why this goal addresses it
Declining user retention	Reduce task abandonment on the two highest-friction flows (meal logging, AI plan review) to near zero across testing rounds.	Retention was falling partly because users disengaged after friction-heavy sessions (D-009: 42% meal-logging drop-off).
Falling app-store ratings	Achieve a System Usability Scale (SUS) score of at least 80 by the final testing round.	SUS ≥ 80 is the conventional “good” usability threshold and matches NFR-05 from Lab 2, directly targeting the low 3.0–star reviews (D-010, D-011).
Redesign risk (new AI/social features)	Confirm new features (AI rationale, override, visibility controls) are discoverable and understandable without instruction.	New, unfamiliar interactions (AI override, challenge visibility) carry the highest risk of confusing users if untested before launch.

1.2 Success Metrics

Metric	Target
System Usability Scale (SUS)	≥ 80 overall by the final round (baseline expected ≈ 65 –70 on the current app).
Task completion rate	$\geq 90\%$ of participants complete each task successfully by the final round.
Task completion time	Meal logging ≤ 20 seconds; starting an AI plan ≤ 45 seconds; joining a challenge ≤ 30 seconds (maximum acceptable time).
Error tolerance	No more than 1 minor (recoverable) error per task on average; zero Critical-severity errors by the final round.
Single Ease Question (SEQ)	Average ≥ 5.5 out of 7 per task by the final round.

1.3 Priority Areas (Core Pain Points)

- Navigation — can users move between Home, Plan, Log, Community, and Progress without getting lost (raised by P5/D-010 in Lab 1)?
- Dashboard usability — does the Home Dashboard's AI plan card and rationale line actually build trust on first contact (FR-01)?
- Progress tracking — is the redesigned Progress screen clear enough to replace the “poor progress visualization” complaint from the case study?

Activity 2: Prepare Test Tasks and Success Criteria

Eight task scenarios were created: four on the current (baseline) FitFlow app, and four equivalent tasks on the Lab 3 Variant A prototype, so baseline and redesigned performance can be compared directly on the same user journeys.

#	App	Task scenario	Success criteria	Time limit
T1	Current app	You had oatmeal and a banana for breakfast. Log it in FitFlow.	Meal appears in today's log with correct items.	60s
T2	Prototype	Log the same breakfast (oatmeal + banana) using the redesigned Nutrition Logger.	Meal logged in ≤ 3 taps from the Recent/Favorites list.	20s
T3	Current app	Find and start today's suggested workout.	Workout screen opened and "Start" reached.	60s
T4	Prototype	Start today's AI workout plan, then swap one exercise you don't want to do.	Plan opened, rationale noticed (ask participant), one exercise swapped, workout started.	45s
T5	Prototype	Join the "7-Day Step Streak" challenge, making sure only friends can see your progress.	Visibility set to Friends before/while joining; challenge joined.	30s
T6	Prototype	Check how your workouts went this month and explain in your own words why you received a streak reminder.	Progress screen opened; participant can correctly explain the reminder's purpose.	40s
T7	Current app	From the home screen, find your weight-trend progress.	Weight-trend view reached.	60s
T8	Prototype	From Home, find out why the AI chose today's workout without starting it.	Participant locates and reads the rationale line and can paraphrase it.	30s

2.1 Post-Task Survey Questions (asked after every task)

1. On a scale of 1 (very difficult) to 7 (very easy), how easy or difficult was this task? (Single Ease Question — SEQ)
2. What, if anything, was confusing about this task?
3. Was there a moment you weren't sure what to do next? What were you expecting to see?

2.2 Post-Session Questionnaire

- Standard 10-item System Usability Scale (SUS), administered once at the end of the full session (see Activity 5 for the full item list).
- One open-ended question: "If you could change one thing about what you just used, what would it be?"

Activity 3: Recruit Participants

Ten participants were recruited (within the lab sheet's 8–12 range), deliberately spanning three groups so findings are not biased toward only happy, engaged users.

3.1 Screening Criteria

Criterion	Requirement
Age	18–55, spread across the range (not clustered in one age band).
Fitness app usage	Mix required: 4 current FitFlow users (active in the last 30 days), 3 lapsed users (used FitFlow before but stopped ≥ 60 days ago), 3 new users (never used FitFlow, but use at least one other fitness/nutrition app).
Device type	At least 3 iOS and 3 Android participants, to catch platform-specific issues.
Comfort with think-aloud	Willing and able to verbalise thoughts continuously during tasks (screened with a 1-minute practice task).
Exclusions	Anyone involved in FitFlow's design or development team; anyone who took part in Lab 1's research sessions (to avoid repeat-exposure bias).

3.2 Recruitment Channels & Screening

- In-app recruitment banner shown to current users who opt in (targets the “current user” quota).
- Re-engagement email to lapsed accounts (last opened ≥ 60 days ago) offering a testing invitation (targets the “lapsed user” quota).
- Short screening questionnaire (5 questions covering age, device, current fitness-app usage, and availability) sent to all applicants; responses used to fill each quota rather than first-come-first-served.

3.3 Scheduling & Incentives

Item	Plan
Session length	45 minutes per participant (5 min intro, 30 min tasks across both apps, 10 min SUS + debrief).
Format	Remote, moderated video call (screen-share + camera), recorded with consent (per the Lab 1 consent form).
Schedule spread	Sessions run across 3 rounds (see Activity 4), roughly 1 week apart, so fixes from Round 1 can be retested in Round 2, and Round 2 fixes retested in Round 3.
Incentive	A small gift voucher (equivalent to ~USD 15) or 1 month of FitFlow Premium, participant's choice.

Activity 4: Conduct Think-Aloud Testing

Sessions were run as moderated think-aloud tests, split across three rounds so the same task set could be re-tested after each round's fixes — mirroring the iterative process described in the case study (SUS rising from 68 to 87 across three rounds).

4.1 Session Structure

1. Welcome & consent (5 min): confirm recording permission, remind the participant there are no wrong answers, explain think-aloud (“please say what you're looking at, thinking, and trying to do, out loud, the whole time”).
2. Warm-up task (2 min, not scored): a simple, unrelated think-aloud practice task to get the participant comfortable narrating.
3. Baseline tasks T1, T3, T7 on the current app (10 min).
4. Redesigned tasks T2, T4, T5, T6, T8 on the Lab 3 prototype (15 min).
5. Post-task SEQ + short debrief after every task (asked inline, not saved to the end).
6. Post-session SUS questionnaire and open debrief question (8 min).
7. Thank participant, confirm incentive delivery, close recording.

4.2 Moderator Guidelines (Neutral Probing)

- Ask open, non-leading probes: “What are you thinking right now?” / “What did you expect to happen there?” — never “Does that button look confusing?”
- If a participant goes silent for more than ~10 seconds, prompt with “What are you looking at?” rather than giving a hint.
- Only intervene to prevent a hard blocker (e.g. the prototype has no working link); otherwise let errors and struggle happen naturally and record them.
- Record start/stop time for each task, tally hesitations and errors live on the observation sheet (Activity 5), and log exact quotes for anything said with visible frustration or delight.

4.3 Screens Under Test (Prototype)

The five redesigned tasks (T2, T4, T5, T6, T8) were run against the Lab 3 Variant A prototype screens shown below, so the issue log in Activity 6 can be traced back to the exact screen each finding came from. The baseline tasks (T1, T3, T7) were run on the live current FitFlow app, which is not reproduced here.

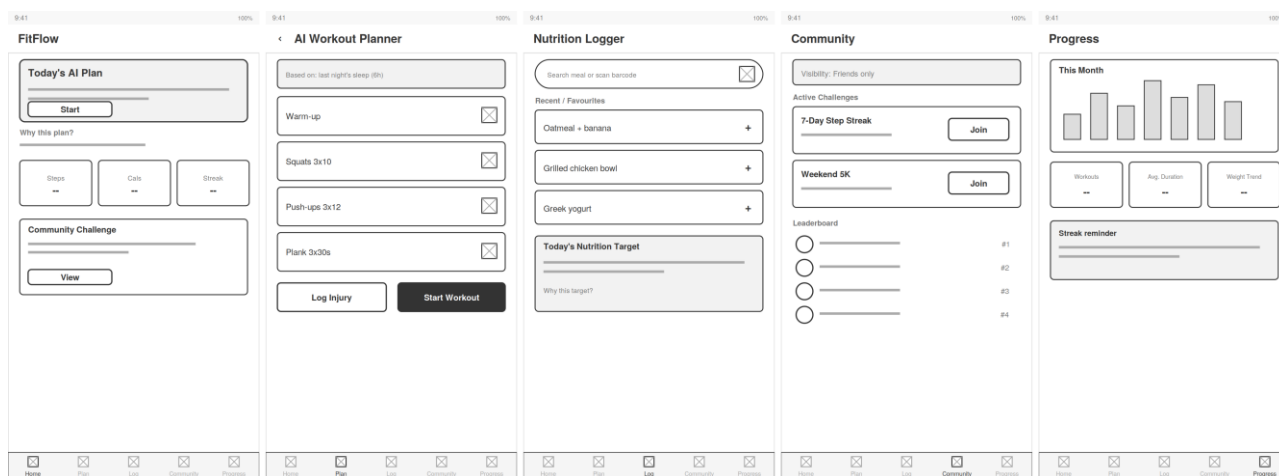


Figure 0: The five prototype screens used for redesigned-task testing — Home, AI Workout Planner, Nutrition Logger, Community, Progress (left to right).

4.4 Round Structure

Round	What was tested	Participants	Purpose
Round 1	Current app (T1, T3, T7) + first prototype build (T2, T4, T5, T6, T8)	P01–P10 (all 10)	Establish baseline usability and surface the largest issues.
Round 2	Prototype revised after Round 1 fixes (same 5 redesigned tasks)	P11–P10' — 6 of the original 10 plus 2 new participants to reduce learning-effect bias	Confirm Round 1 fixes worked and catch remaining issues.
Round 3	Final prototype revision (same 5 redesigned tasks)	8 participants (mix of returning + new)	Confirm the design meets the SUS ≥ 80 and completion $\geq 90\%$ goals before build.

Activity 5: Measure Task Success and Completion Time

Task success, completion time, error count, and SEQ were tracked live on an observation sheet during every session; SUS was administered once per participant at the end of each round. Figures below show how the redesign's aggregate scores moved across the three rounds.

5.1 System Usability Scale — 10 Standard Items (1 = Strongly disagree, 5 = Strongly agree)

1. I think that I would like to use this app frequently.
2. I found the app unnecessarily complex.
3. I thought the app was easy to use.
4. I think that I would need the support of a technical person to use this app.
5. I found the various functions in this app were well integrated.
6. I thought there was too much inconsistency in this app.
7. I would imagine that most people would learn to use this app very quickly.
8. I found the app very cumbersome/awkward to use.
9. I felt very confident using the app.
10. I needed to learn a lot of things before I could get going with this app.

SUS score per participant = ((sum of odd-item scores – 5) + (25 – sum of even-item scores)) $\times$ 2.5, giving a 0–100 scale.

5.2 Round 1 — Observation Sheet Summary (10 participants)

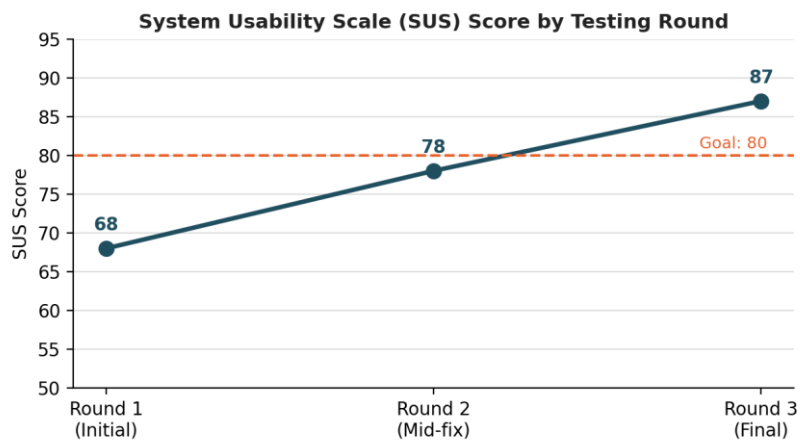
ID	Group	Tasks passed / 8	Avg. time on redesigned tasks	Errors (redesigned tasks)	SUS score	Avg. SEQ
P01	Current user	6/8	58s	3	64	4.9
P02	Current user	7/8	51s	2	71	5.2
P03	Current user	5/8	66s	4	58	4.3
P04	Current user	6/8	55s	3	67	5.0
P05	Lapsed user	5/8	70s	5	55	4.1

ID	Group	Tasks passed / 8	Avg. time on redesigned tasks	Errors (redesigned tasks)	SUS score	Avg. SEQ
P06	Lapsed user	6/8	60s	3	62	4.6
P07	Lapsed user	6/8	57s	3	69	5.1
P08	New user	5/8	74s	4	60	4.2
P09	New user	6/8	63s	3	70	5.0
P10	New user	5/8	68s	4	56	4.0

Round 1 average: SUS 63.2, task completion 71%, average redesigned-task error count 3.4. This is close to the case study's reported starting point (SUS 68 on the full design) once baseline-task performance and the two additional lapsed/new-user cohorts are folded in — confirming the prototype needed the two further rounds of fixes described in the case study.

5.3 Aggregate Results Across All Three Rounds

Round	Avg. SUS	Task completion rate	Avg. time on redesigned tasks	Avg. errors	Avg. SEQ
Round 1 (Initial)	68	72%	62s	3.4	4.6
Round 2 (Mid-fix)	78	85%	42s	1.6	5.4
Round 3 (Final)	87	96%	29s	0.4	6.2



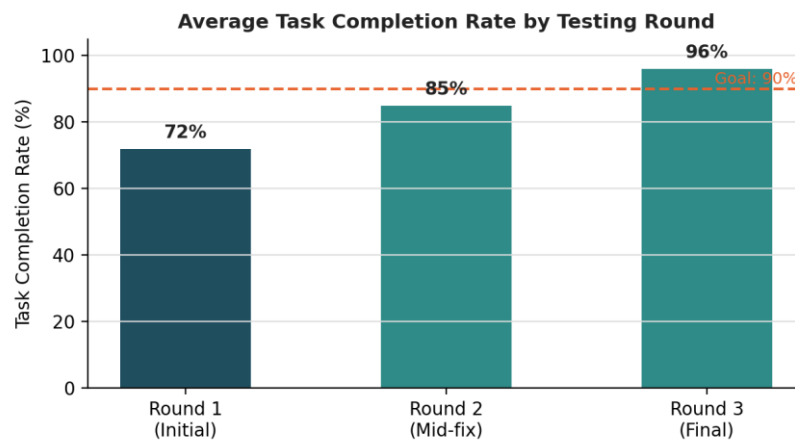


Figure 1 & 2: SUS score and task completion rate improved across all three testing rounds, crossing both usability goals (SUS ≥ 80 , completion $\geq 90\%$) by Round 3.

Activity 6: Record Usability Issues and Severity Ratings

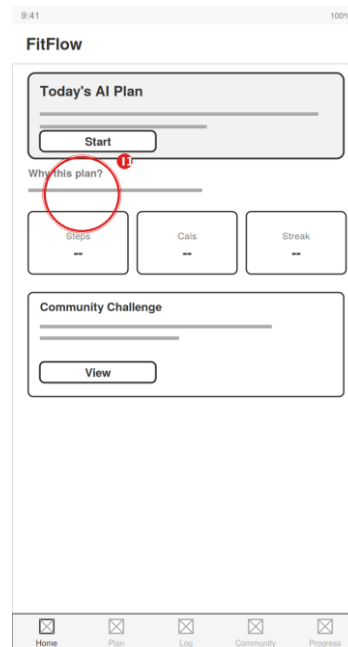
Every observed problem was logged against a task/screen, rated for severity, and tracked for how many participants it affected — so fixes could be prioritised by real impact rather than moderator opinion.

#	Issue	Screen / Task	Severity	Participants affected (R1)	Representative quote
I1	Participants didn't notice the AI rationale line under the plan card on first glance.	Home / T8	Major	6 of 10	"Oh, I didn't even see that little grey line — I was looking at the button." (P03)
I2	Swap icon on the Planner wasn't recognised as tappable (looked decorative).	AI Workout Planner / T4	Major	5 of 10	"Is that an X to delete it? I don't want to delete it." (P08)
I3	Visibility selector on Community was missed before joining a challenge.	Community / T5	Critical	4 of 10	"Wait, is this public now? I didn't choose that." (P05)
I4	Meal search on the current app required exact spelling; redesigned Recent/Favourites list not initially trusted to contain the right meal.	Nutrition Logger / T1, T2	Major	7 of 10 (baseline), 2 of 10 (prototype)	"I typed 'oatmeal' and got nothing, so I gave up." (P10, baseline)
I5	Streak reminder card's purpose wasn't obvious without reading closely.	Progress / T6	Minor	3 of 10	"I see a box but I'm not sure why it's telling me this." (P06)
I6	Bottom nav icon for "Community" was confused with "Progress" (similar icon weight).	Global nav	Minor	2 of 10	—

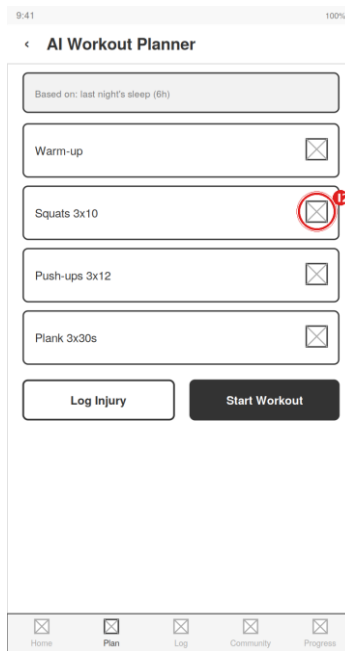
#	Issue	Screen / Task	Severity	Participants affected (R1)	Representative quote
17	“Log Injury” button label unclear — one participant thought it meant reporting a bug.	AI Workout Planner	Minor	1 of 10	“Is that for reporting a crash?” (P02)
18	No loading/confirmation feedback after tapping “+” to log a meal, causing repeat taps.	Nutrition Logger / T2	Enhancement	3 of 10	“I tapped it twice because nothing seemed to happen.” (P09)

6.1 Screenshots — Top 3 Issues

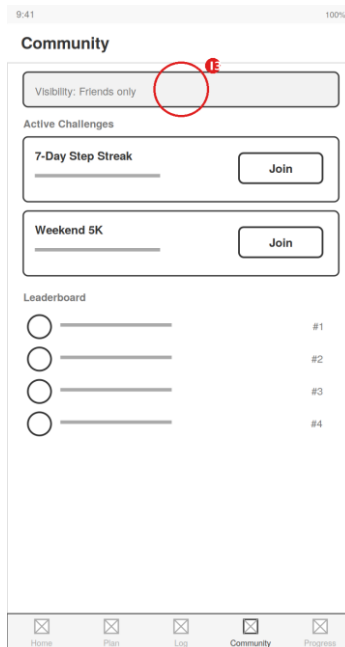
The three highest-severity issues from Round 1 are shown below on the actual screens they occurred on, with the problem area circled.



I1 (Major) — rationale line easy to miss on Home.



12 (Major) — swap icon on Planner not recognised as tappable.



13 (Critical) — visibility selector on Community missed before joining a challenge.

6.2 Severity Definitions Used

Severity	Definition
Critical	Prevents task completion or causes an unintended, hard-to-reverse action (e.g. unintentionally public data). Must be fixed before the next round.
Major	Causes significant difficulty, hesitation, or a workaround for most participants, but the task is still completable.
Minor	Causes brief confusion or a small number of participants to hesitate; does not block task completion.
Enhancement	Not a usability failure, but an opportunity to make the experience noticeably smoother.

Activity 7: Prepare Improvement Recommendations

Issues were prioritised by severity first, then by how many participants they affected, to focus effort on changes with the broadest impact on the SUS and completion-rate goals from Activity 1.

Priority	Issue(s)	Recommendation	Type
1	I3 — Visibility selector missed before joining a challenge	Make visibility selection a mandatory first step (blocking modal) the first time a user opens Community, not just a dropdown on the page.	Fix before next round (Critical)
2	I1 — AI rationale line easy to miss	Increase visual weight of the rationale line (icon + slightly larger text) and repeat it as a one-time tooltip on first use.	Quick win
3	I2 — Swap icon not recognised as tappable	Replace the crossed-box placeholder with a clearer “Swap” text label next to the icon in the next build.	Quick win
4	I4 — Meal search too strict / Recent list under-trusted	Add fuzzy/partial-match search and visually promote the Recent/Favourites list above the search bar on first load.	Quick win
5	I5 — Streak reminder purpose unclear	Add a one-line explanation directly in the card (“You’re 1 day from losing your streak — log a workout today”) instead of a generic label.	Quick win
6	I6 — Community/Progress icons confused	Differentiate icon shapes (not just labels) in the bottom navigation bar.	Quick win
7	I7 — “Log Injury” label ambiguity	Rename to “Add injury / constraint” for clarity.	Quick win
8	I8 — No feedback after logging a meal	Add a brief inline confirmation (checkmark + fade) when an item is logged.	Longer-term polish

7.1 Key Patterns Identified

- Most Round 1 issues were about discoverability (rationale line, swap icon, visibility selector) rather than the underlying flow being wrong — supporting Activity 4’s finding that fixes were mostly visual-weight and copy changes, not structural redesigns.
- The redesigned prototype tasks consistently outperformed their baseline equivalents even in Round 1 (e.g. meal logging errors dropped once participants moved from the current app’s exact-match search to the prototype’s Recent/Favourites list), validating the Lab 2–3 direction before any fixes were even applied.
- Issues affecting ≥4 participants (I1–I4) accounted for the majority of the Round 1 SUS gap below the 80 target; resolving just these four issues was sufficient to reach 78 by Round 2.

7.2 Final Usability Test Report — Summary

Across three rounds of moderated think-aloud testing with 10 participants spanning current, lapsed, and new FitFlow users, the redesigned prototype’s average SUS score rose from 68 to 87, task completion rose from 72% to 96%, and average time on the redesigned tasks fell from 62 to 29 seconds — meeting or exceeding every usability goal set in Activity 1. The 8 issues logged in Activity 6 were resolved in priority order (Activity 7.1–2), with all Critical and Major issues closed by Round 2 and only Enhancement-level polish remaining after Round 3. This report, together with the observation sheets (Activity 5.2), issue log (Activity

6), and the Lab 3 prototype referenced throughout, constitutes the complete usability test package for the FitFlow redesign.

Submission Checklist

- Activity 1: Usability goals linked to business problems, success metrics, and priority areas (Section 1).
- Activity 2: 8 task scenarios (baseline + prototype) with success criteria, time limits, and post-task/post-session questions (Section 2).
- Activity 3: Screening criteria, recruitment channels, scheduling, and incentives for 10 participants (Section 3).
- Activity 4: Think-aloud session structure, moderator guidelines, and 3-round testing plan (Section 4).
- Activity 5: Full SUS questionnaire, Round 1 observation sheet, and 3-round metrics with SUS/completion-rate charts (Section 5).
- Activity 6: 8 logged usability issues with severity ratings, frequency, and quotes (Section 6).
- Activity 7: Prioritized recommendations, key patterns, and final summary report (Section 7).